Protracted Bacterial Bronchitis

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Definition

The current criteria for diagnosing protracted bacterial bronchitis (PBB) are as follows (PBB Clinical):

- ☑ Wet cough lasting at least 4 weeks
- ☑ Absence of other findings to identify another cause of the cough
- ☑ No evidence of an alternative diagnosis after a standard evaluation
- ☑ Resolution of the cough with at least 2 weeks of appropriate antibiotic.

PBB extended: Severe disease, cough resolves after an extended course (4 weeks) of antibiotics.

PBB recurrent: >3 episodes of PBB in 1 year.

Etiology

- ☑ *Streptococcus pneumoniae, Haemophilus influenzae* [mostly nontypable (NTHi)] and *Moraxella catarrhalis* were the main bacteria detected in BAL specimens.
- ☑ There is no certain evidence of the role of viruses in PBB pathogenesis even though different types of viruses have been detected in BAL from children with PBB.
- ☑ There is no correlation with any type of immunodeficiency.

Clinical Features

- ☑ It can occur in any age but common in young children who are otherwise healthy.
- ☑ Systemic symptoms are generally absent.
- ☑ Night-time coughing, shortness of breath during physical exercise, wheeze and exacerbations with upper respiratory tract infections (URTIs) can be presenting symptom.
- ☑ It can coexist with asthma or airway malacia.
- ☑ Tiredness due to disturbed sleep or chronic infection.
- ☑ Usually, no signs on chest examination, occasional crackles and wheeze.
- ☑ It rattles due to airway secretions.

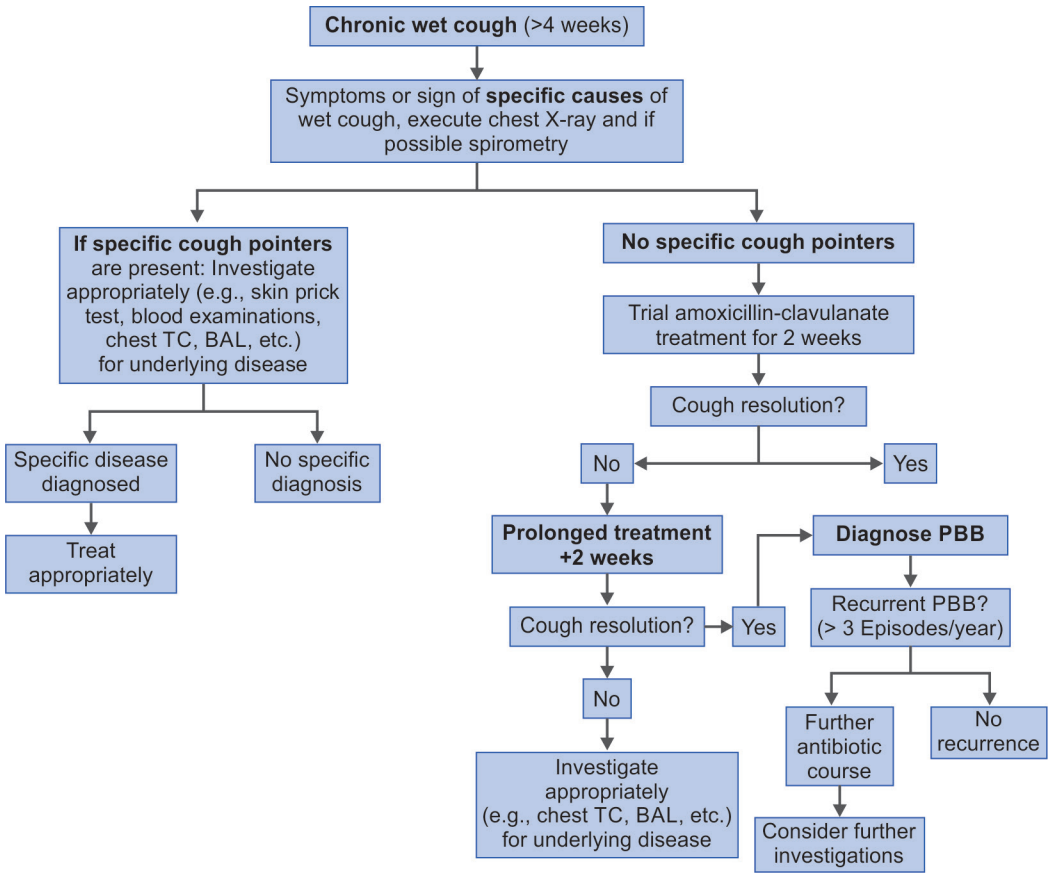
Diagnosis

- ☑ Mainly clinical on the basis of history of all diagnostic criteria of PBB present.
- ☑ *Examination:* Well looking child with normal physical examination. Sometimes wheezing may be associated with PBB.
- ☑ A chest X-ray may be done to exclude other causes, but often does not show any specific finding.
- ☑ If the child is old enough to do lung function testing, a pattern of airway obstruction may be seen.
- ☑ *Bronchoscopy:* Purulent airway secretions and large airway malacia are common findings.

Differential Diagnosis

- ☑ Forgotten foreign body
- ☑ Airway congenital malformations
- ☑ Recurrent aspirations
- ☑ *Muco-ciliary disorders:* Cystic fibrosis/primary ciliary dyskinesia (PCD)
- ☑ Immunodeficiencies
- ☑ Asthma or recurrent wheezing.

Approach to chronic wet cough.

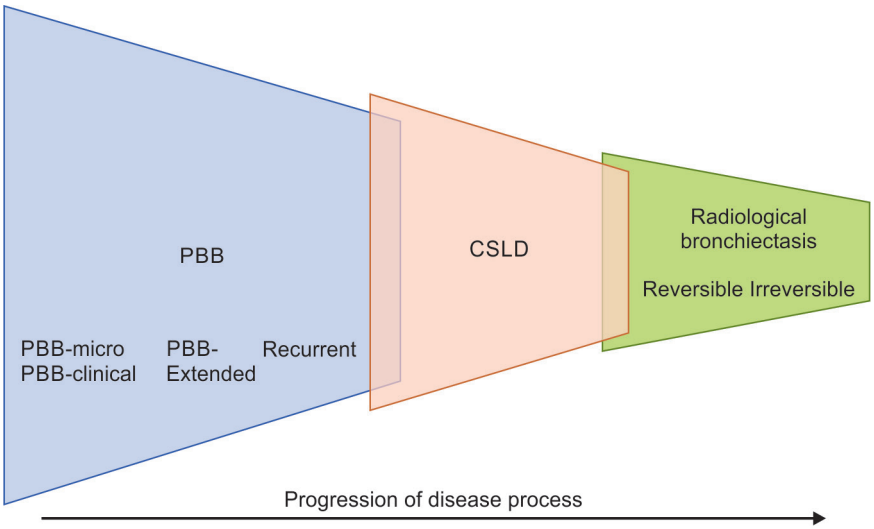


Treatment

- ☑ The antibiotic of choice is amoxicillin-clavulanate in conventional dosage. The minimum duration of oral antibiotics required was found to be 2 weeks. Treatment lasting <2 weeks has a risk for relapse.
- ☑ *Alternatives:* Oral 2nd or 3rd generation cephalosporins, trimethoprim/sulfamethoxazole (TMP-SMX) or macrolide: Lacks demonstrable efficacy.
- ☑ Recurrences are common and it is reasonable to wait for 2 weeks to see if these are related to a new viral infection.
- ☑ Recurrences should be treated with 2 weeks of antibiotics.
- ☑ Children with recurrent PBB (>3 episode/year) need to be treated with 6 weeks of antibiotics.
- ☑ Children with bronchoscopic evidence of PBB (Purulent secretions, heavy growth of typical bacteria ccf >10^4, neutrophilia >6%) should be treated with 4 weeks of antibiotics, guided by culture and sensitivity results.
- ☑ For children who do not respond to 4 weeks of antibiotic treatment or have more than one episode require further investigations (bronchoscopy, sweat chloride test, immuno-deficiency work up).

Natural Course of PBB

- ☑ Protracted bacterial bronchitis which is not treated adequately can predispose to bronchiectasis and chronic suppurative lung disease (CSLD), thus exhibiting a single clinical continuum from PBB to CSLD.
- ☑ Protracted bacterial bronchitis typically responds to a 2–4-week course of appropriate antibiotics. However, in cases of recurrent episodes or a poor response to 4 weeks of antibiotics, the child needs to be investigated for other causes of chronic wet cough and the possibility of CSLD.



Conclusion

- ☑ Protracted bacterial bronchitis is an underdiagnosed entity, albeit one of the most common causes for chronic wet cough in children.
- ☑ The clinical criteria for the diagnosis of PBB is proven to be sufficient to warrant therapy.
- ☑ The treatment of choice is a 2-week course of co-amoxiclav in conventional doses.
- ☑ Protracted bacterial bronchitis when left untreated can predispose to bronchiectasis or CSLD and hence needs to be diagnosed and treated early.
- ☑ Children with recurrent PBB would require a bronchoscopic evaluation of airway, CT scan, sweat test and immunodeficiency work-up to evaluate for other causes of chronic wet cough and may need up to 6 weeks of antibiotic therapy.

Further Reading

- ☑ Armstrong D. Long-term follow-up of children with protracted bacterial bronchitis: some answers, and more questions. Respirology. 2021;26:218-9.
- ☑ Chang AB, Upham JW, Masters IB, Redding GR, Gibson PG, Marchant JM, et al. Protracted bacterial bronchitis: the last decade and the road ahead. Pediatr Pulmonol. 2016;51:225-42.
- ☑ Di Filippo P, Scaparrotta A, Petrosino MI, Attanasi M, Di Pillo S, Chiarelli F, Mohn A. An underestimated cause of chronic cough: the protracted bacterial bronchitis. Ann Thorac Med. 2018;13:7-13.
- ☑ Kantar A, Chang AB, Shields MD, Marchant JM, Grimwood K, Grigg J, et al. ERS statement on protracted bacterial bronchitis in children. Eur Respir J. 2017;24;50:1602139.
- ☑ Reddy KB, Gill KS, Nair S, Bhattacharya BG. Protracted bacterial bronchitis: an underdiagnosed cause for chronic wet cough in children. Pediatr Inf Dis. 2020;2(1):19-22.
- ☑ Ruffles TJC, Marchant JM, Masters IB, Yerkovich ST, Wurzel DF, Gibson PG, et al. Outcomes of protracted bacterial bronchitis in children: a 5-year prospective cohort study. Respirology. 2021;26:241-8.

Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Kawasaki Disease

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